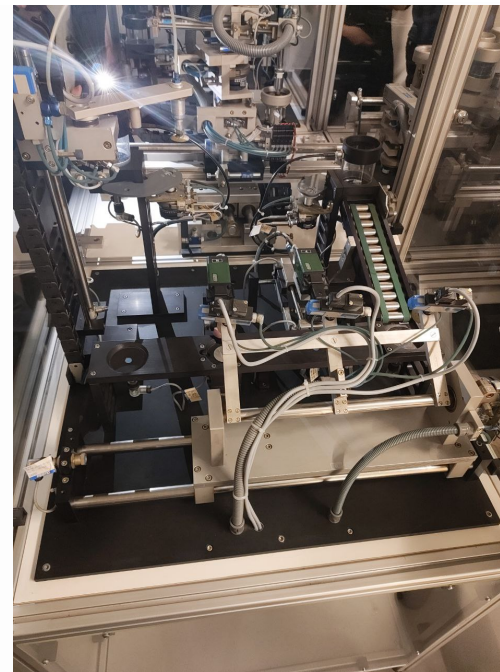




Company / customer Riga Technical University (RTU), Mechatronics
 Project description FESTO station — Loading / Processing /
 Job number RTU-MPS-001
 Commission RTU Mechatronics

Project name Festo Station
 Responsible for project Deniss Kašins, Daniels Skots Lavs, Ruslans Novikovs
 Last EPLAN version used 2026.0.3

 Manufacturing date 2025
 Type Mechatronic station, pneumatic (FESTO MPS: Zones 1–4)
 Place of installation RTU, Ķīpsala campus
 Power supply 230 V~ 50 Hz, 10 A; Pneumatics 6 bar (components up to 10 bar)
 Input lead H05VV-F 3×1.5 mm² (mains); 0.34–0.75 mm² (24 V DC control)
 Control voltage 24 V DC
 Special customer regulations No



Created on 05.06.2025
 Edit date 28.01.2026 by (short name) dino3

Number of pages 27

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TECHNICAL IMPLEMENTATION REGULATIONS



WIRING REGULATIONS

(Symbol in accordance with IEC 60757)

WIRING COLORS

Main current AC	Black (BK)	Control circuit AC > 50V	Red (RD)	External voltage	Orange (OG)
Main current DC	Black (BK)	Control circuit AC < 50V	Red (RD)	Interlock circuit	Orange (OG)
Neutral conductor	Blue (BU)	Control circuit DC	Gray (GY)	Measuring lines	Violet (VT)
Protective wire	Green-yellow (GNYE)			Current transformer	Brown (BN)

MINIMUM CROSS-SECTIONS

Main current	2,5 mm ²
Protective wire	1,5 mm ²
Control line	1,5 mm ²
PLC wiring	0,75 mm ²

WIRING

0.5 - 1.0 mm ²	H05V-K PVC conductor line
1.5 - 150 mm ²	H07V-K PVC conductor line

MATERIAL REGULATIONS

ENCLOSURES

Enclosures	RITTAL VX
Enclosure base	RITTAL VX
Enclosure color	RAL 9016
Degree of protection	IP43

TERMINALS

to 2.5 mm ²	Terminal blocks Phoenix
4 to 150 mm ²	Terminal blocks Phoenix

GENERAL SAFETY REGULATIONS



During plant operation certain items are under dangerous voltage! Non-observance of safety instructions can result in death, serious injuries or material damage.

Only specialist personnel may carry out transport, installation and commissioning work.

The applicable valid standards as well as the national and / or company-specific accident prevention regulations are to be observed.

The following safety instructions are to be observed:

The erection, commissioning, troubleshooting as well as repair of the plant may only be carried out by qualified personnel that is familiar with the corresponding operating instructions.

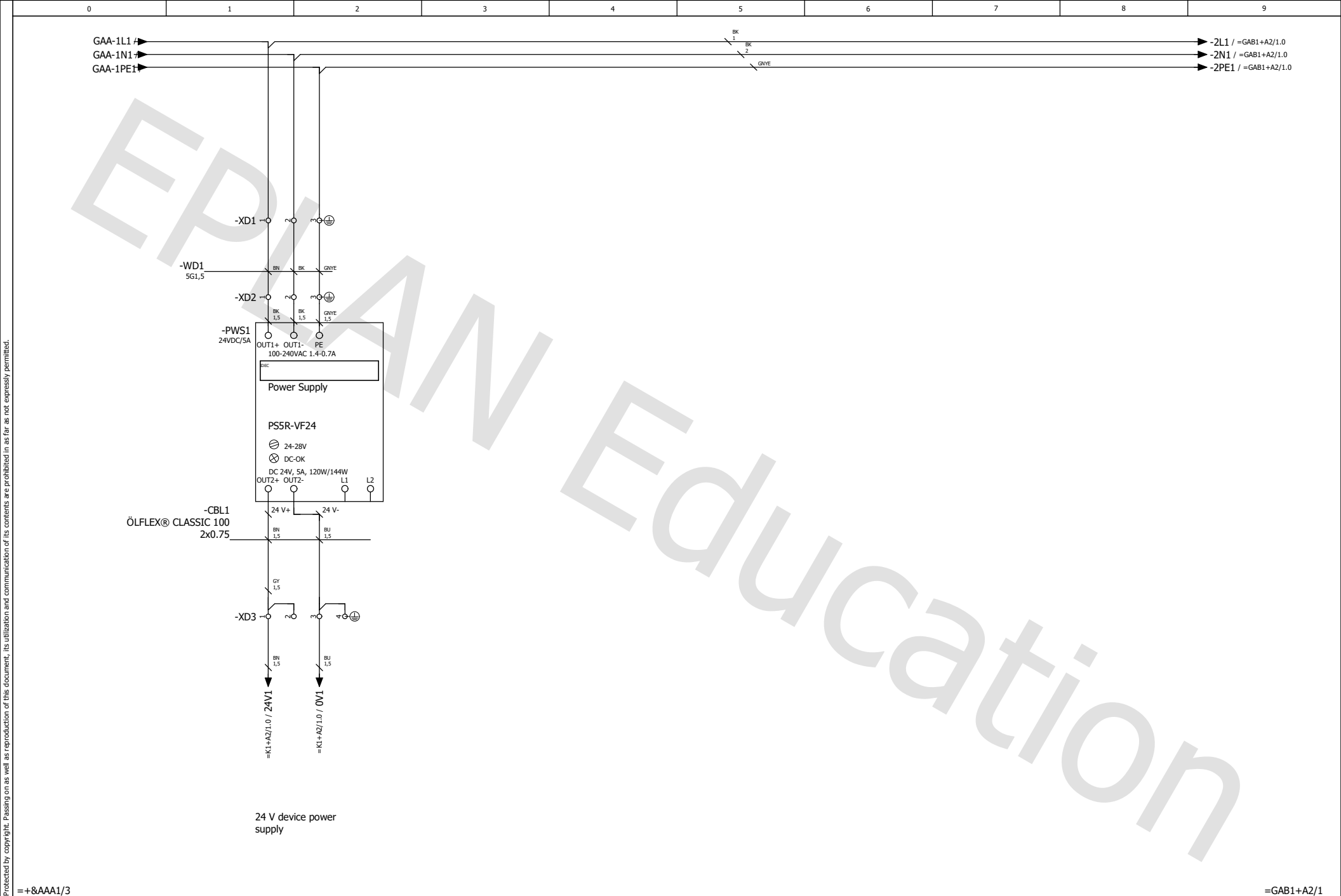
Mounting of the devices has to be effected in accordance with the valid standards, state and local regulations. Proper grounding and conductor dimensioning as well as proper short-circuit proofing have to be ensured. These measures serve to ensure the safety of the plant and of the operating personnel.

Before carrying out safety checks, maintenance work and repair measures ensure that all the power supplies are switched off, are secured against being switched on unintentionally and are marked correspondingly.

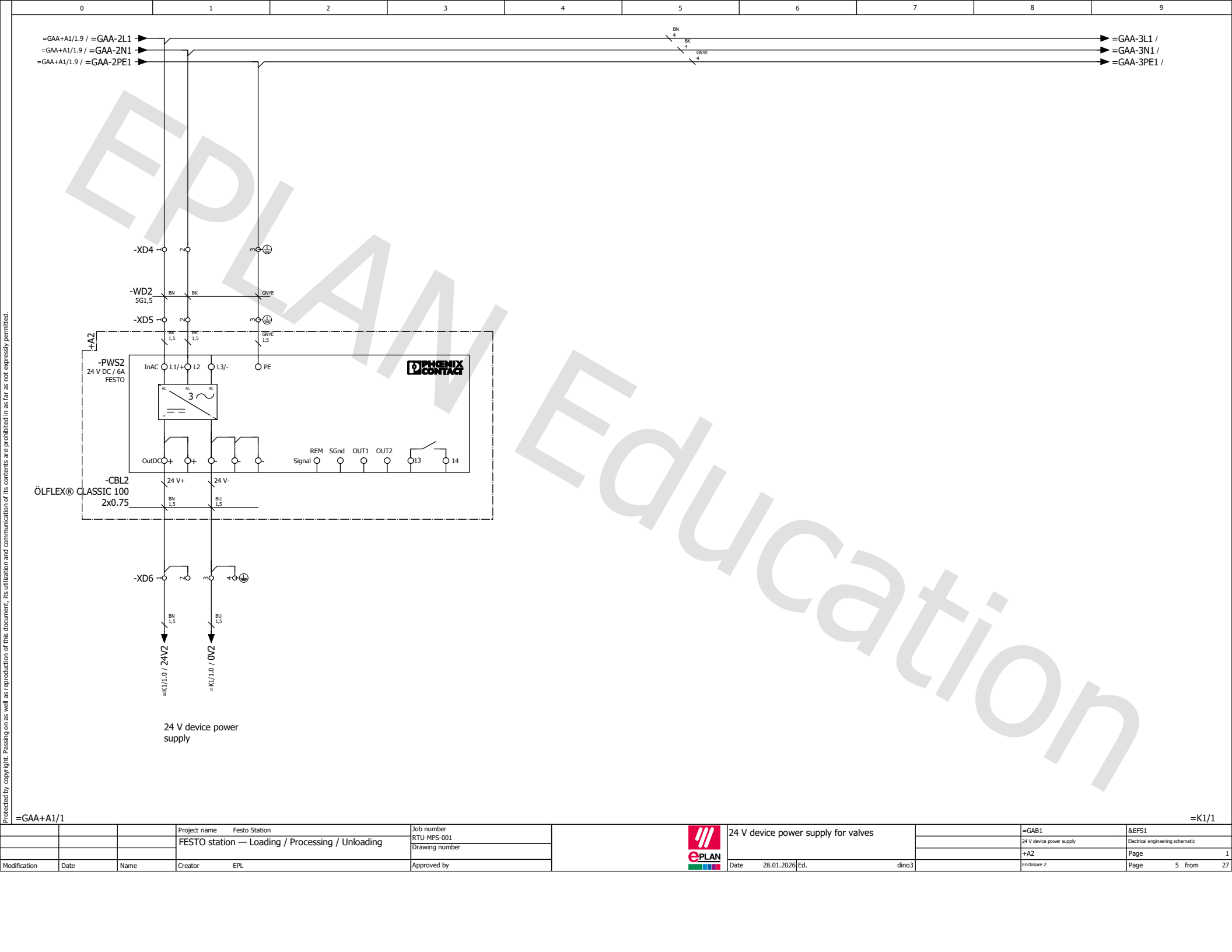
Only test devices that are in a technically perfect condition and are suitable for the respective measurement may be used to carry out measurements!

The instructions specified in the respective operating instructions are to be followed strictly!
It is mandatory that hazard, warning and safety instructions be observed!

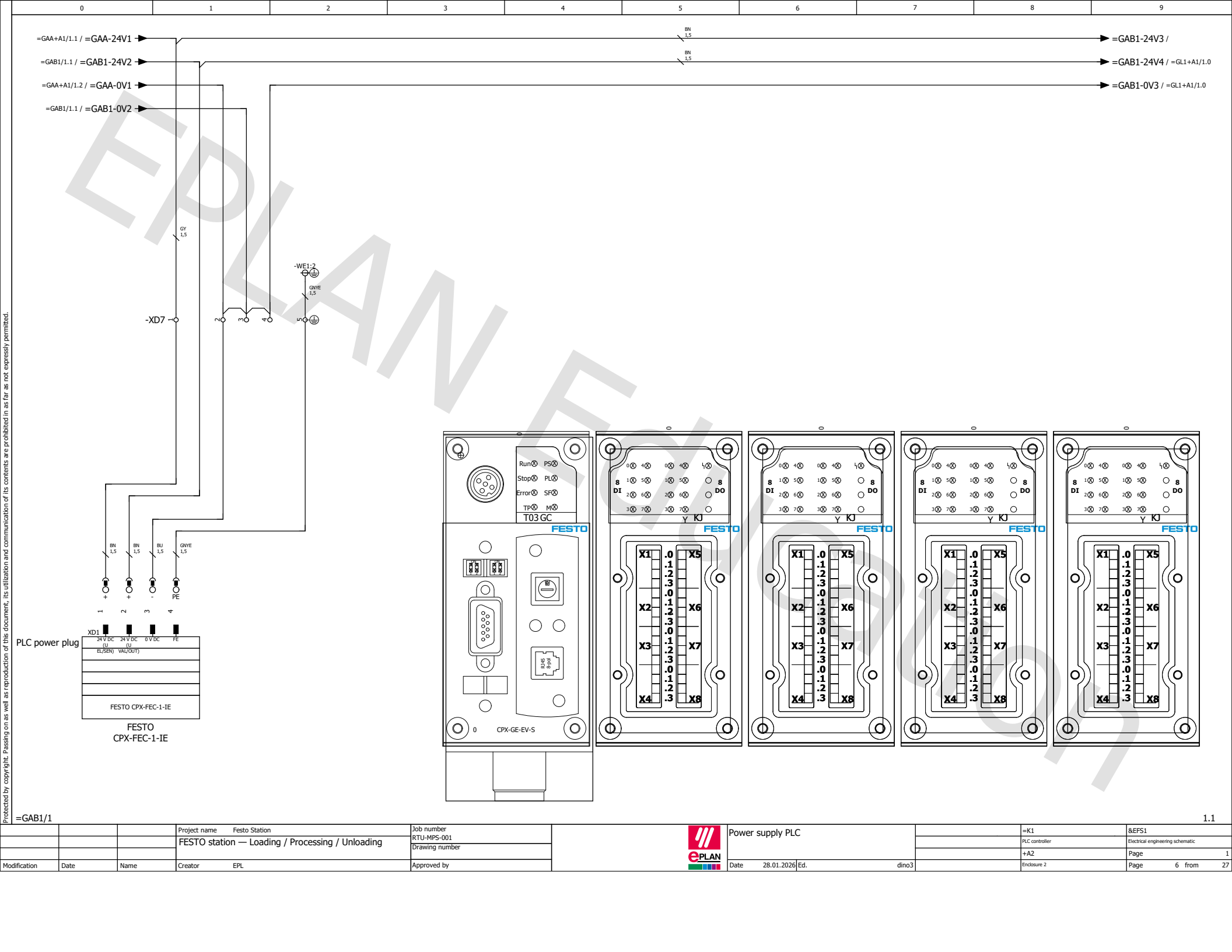
All doors and covers are to be kept closed during the plant operation.
If cooling devices are installed in the plant, ensure that these systems operate trouble-free.
These include the regular cleaning of the filters, in as far as they exist.

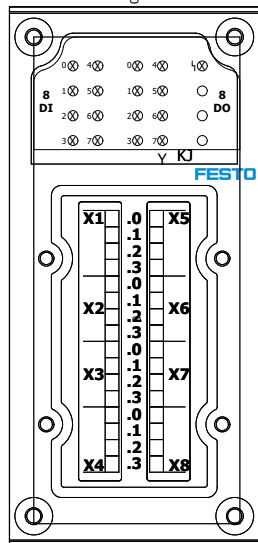


			Project name Festo Station		Job number RTU-MPS-001			24 V device power supply for sensors		=GAA	&EFS1	
			FESTO station — Loading / Processing / Unloading		Drawing number					400 V power supply	Electrical engineering schematic	
										+A1	Page 1	
Modification	Date	Name	Creator EPL		Approved by		Date 28.01.2026		Ed. dino3	Enclosure 1		
										Page 4 from 27		



			Project name	Festo Station	Job number	 24 V device power supply for valves		=GAB1	&EFS1
			FESTO station — Loading / Processing / Unloading		RTU-MPS-001			24 V device power supply	Electrical engineering schematic
					Drawing number			+A2	Page 1
Modification	Date	Name	Creator	EPL	Approved by	Date 28.01.2026	Ed. dino3	Enclosure 2	Page 5 from 27

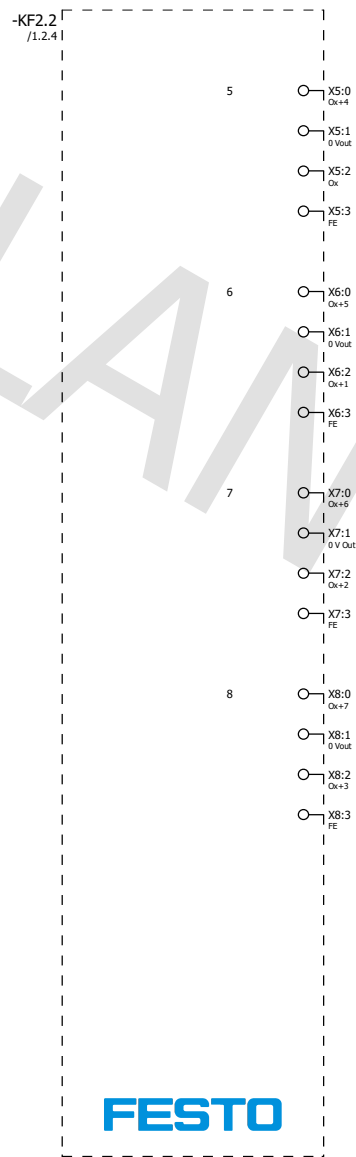




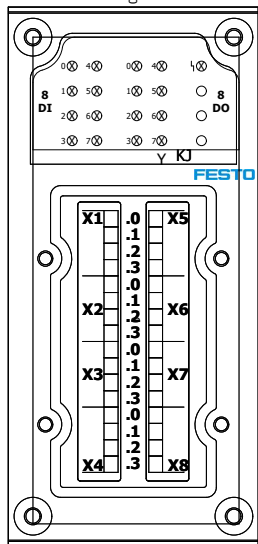
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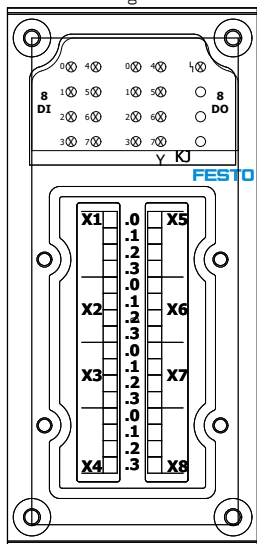
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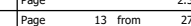


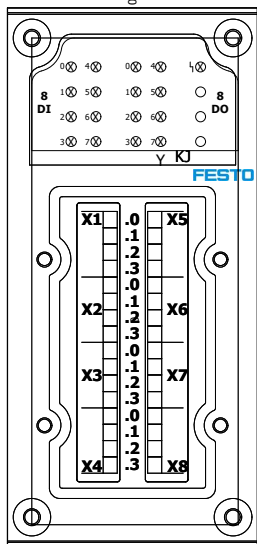
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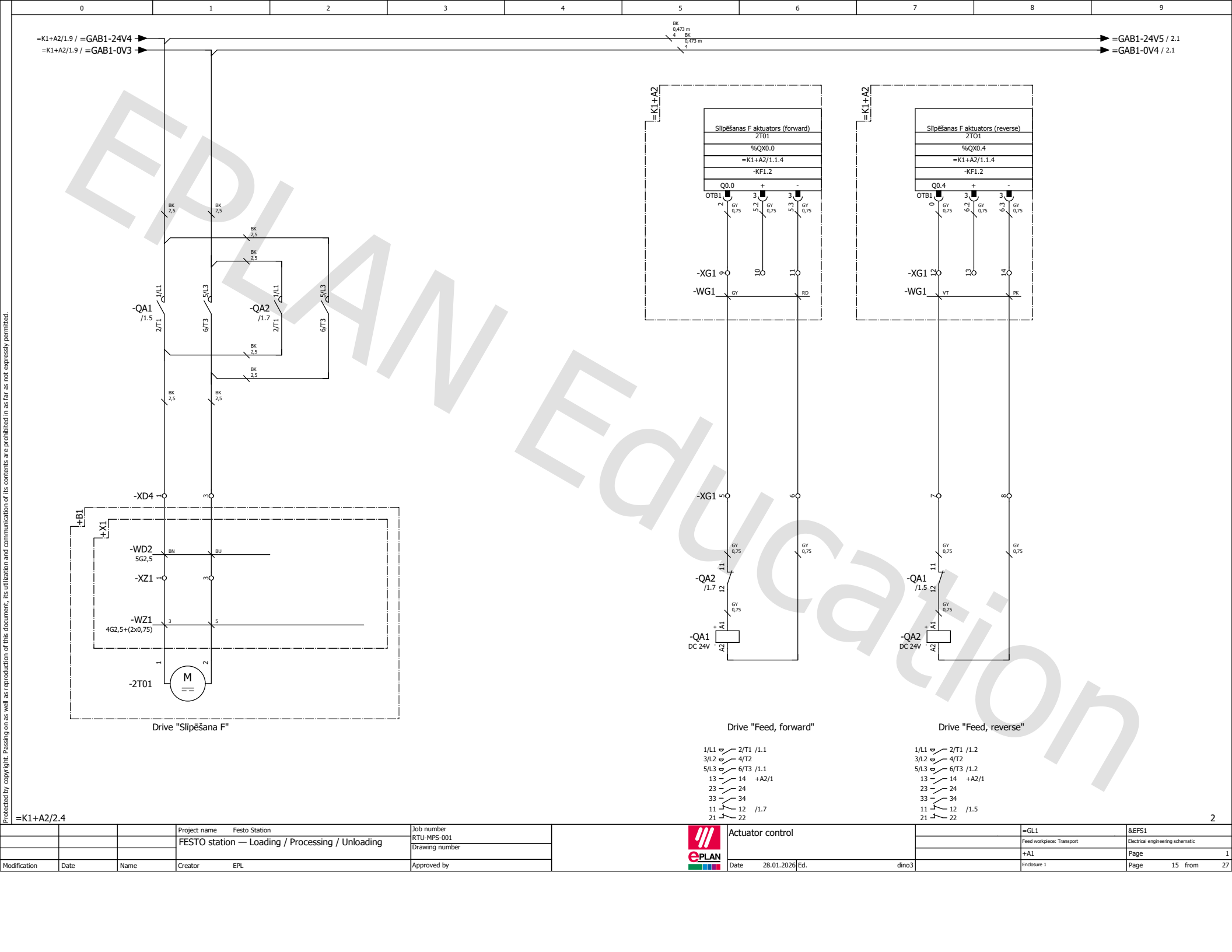


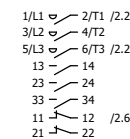
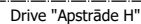


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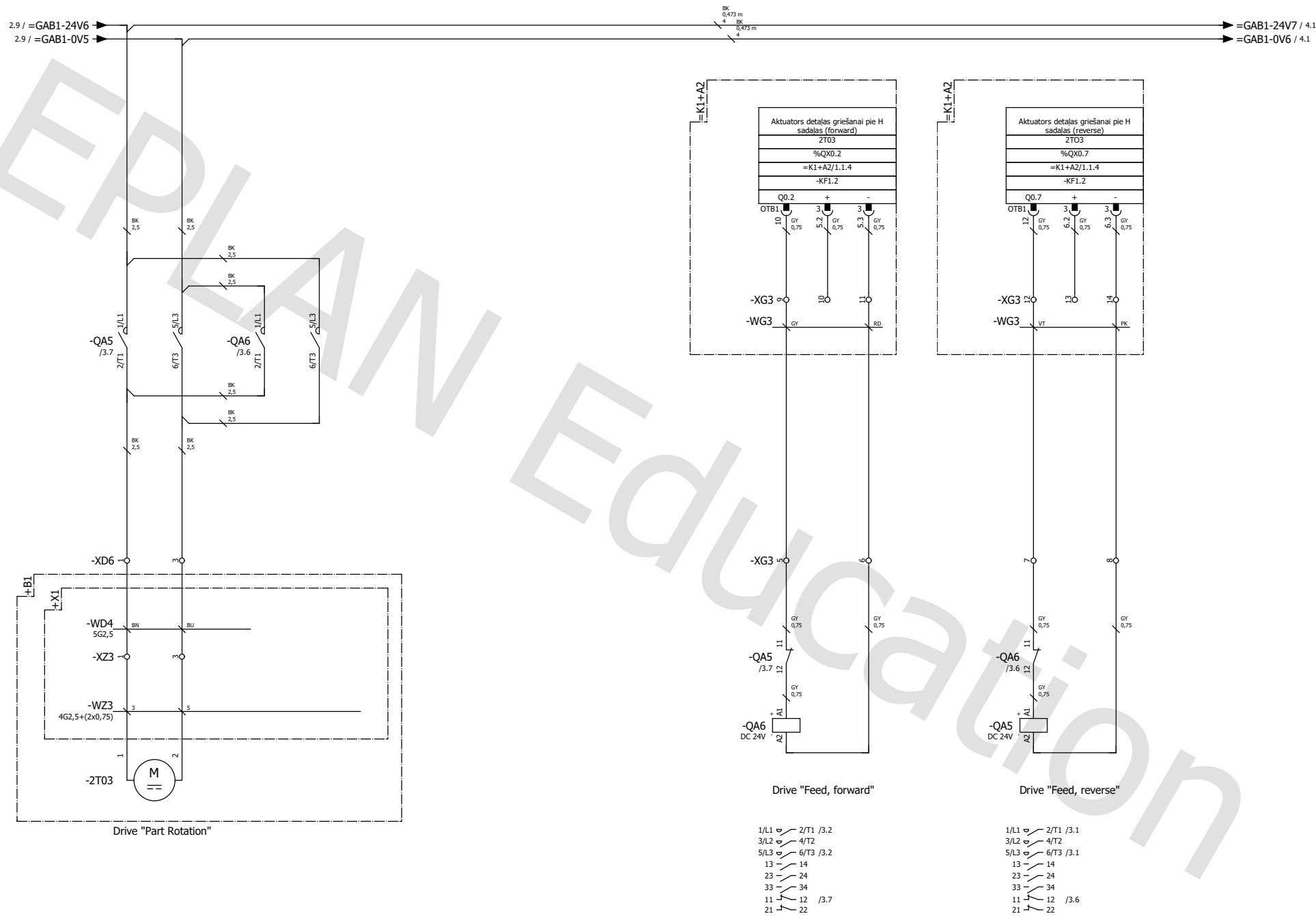



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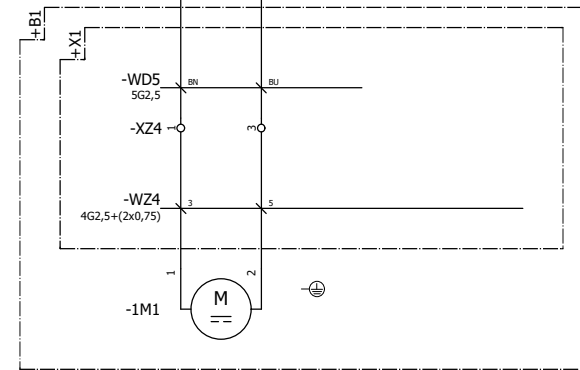




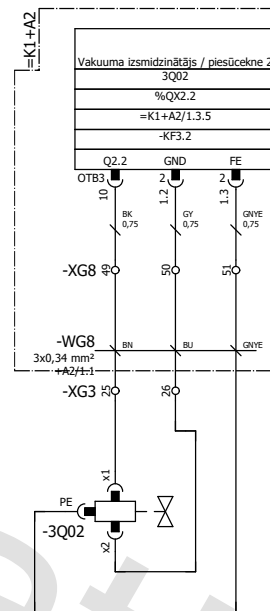
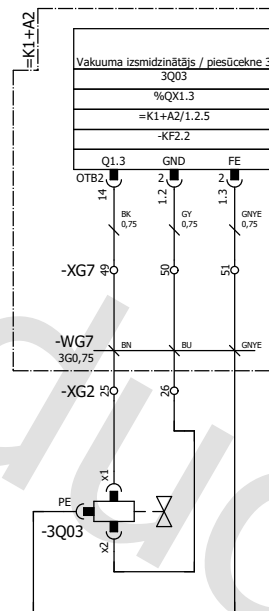
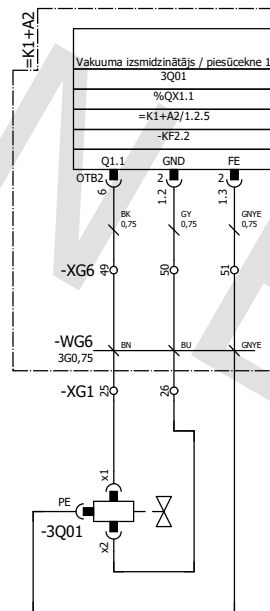
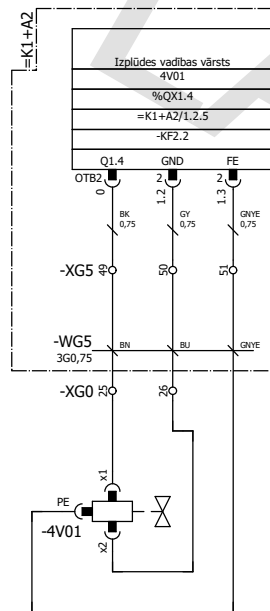
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			FESTO station — Loading / Processing / Unloading			Drawing number				Feed workpiece: Transport	Electrical engineering schematic				
										+A1	Page	3			
Modification	Date	Name	Creator	EPL	Approved by		Date		28.01.2026	Ed.	dino3	Enclosure 1	Page	17	from

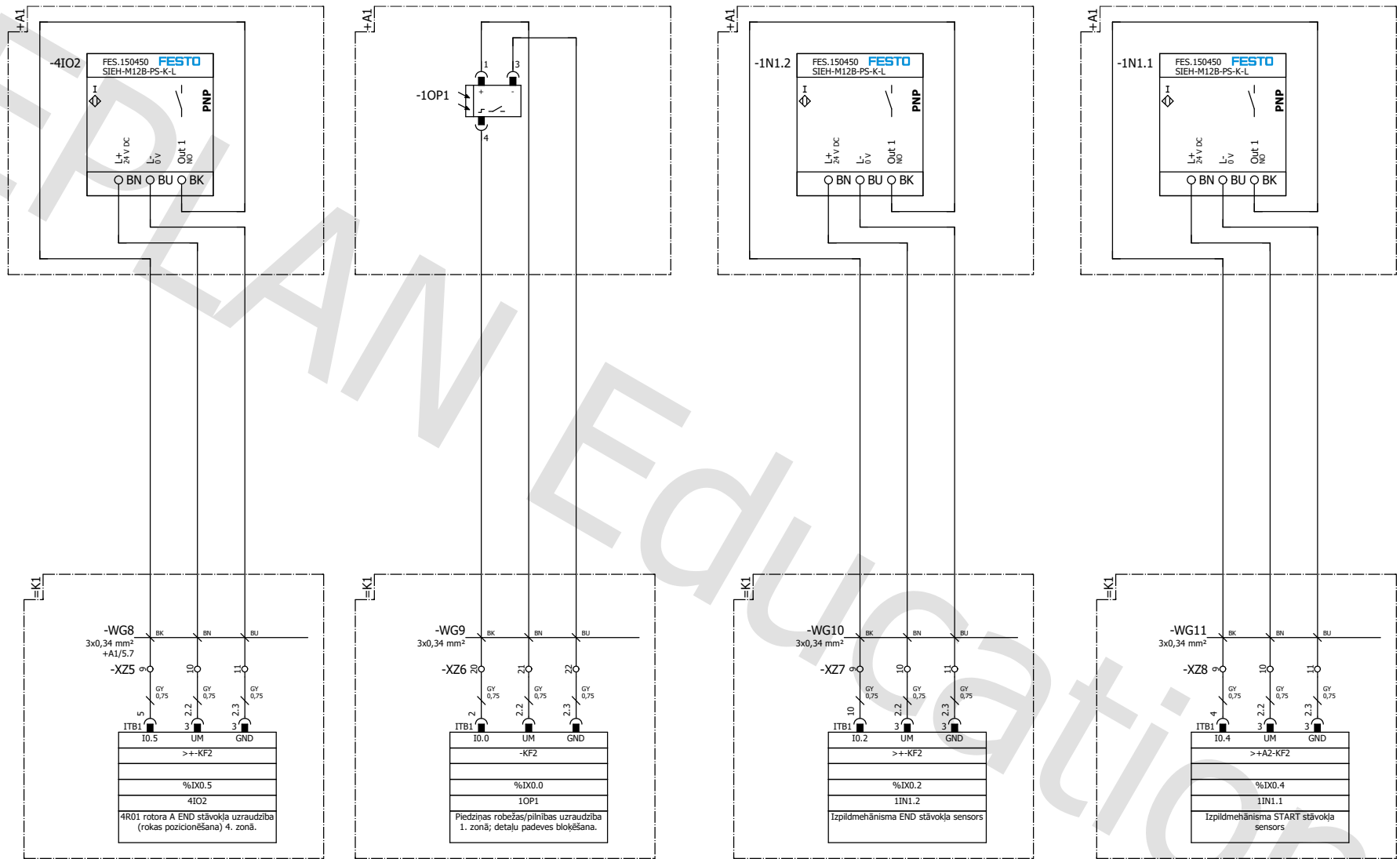


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					Drawing number							4			
Modification	Date	Name	Creator	EPL	Approved by		Date	28.01.2026	Ed.	dino3	Enclosure 1	Page	18	from	27



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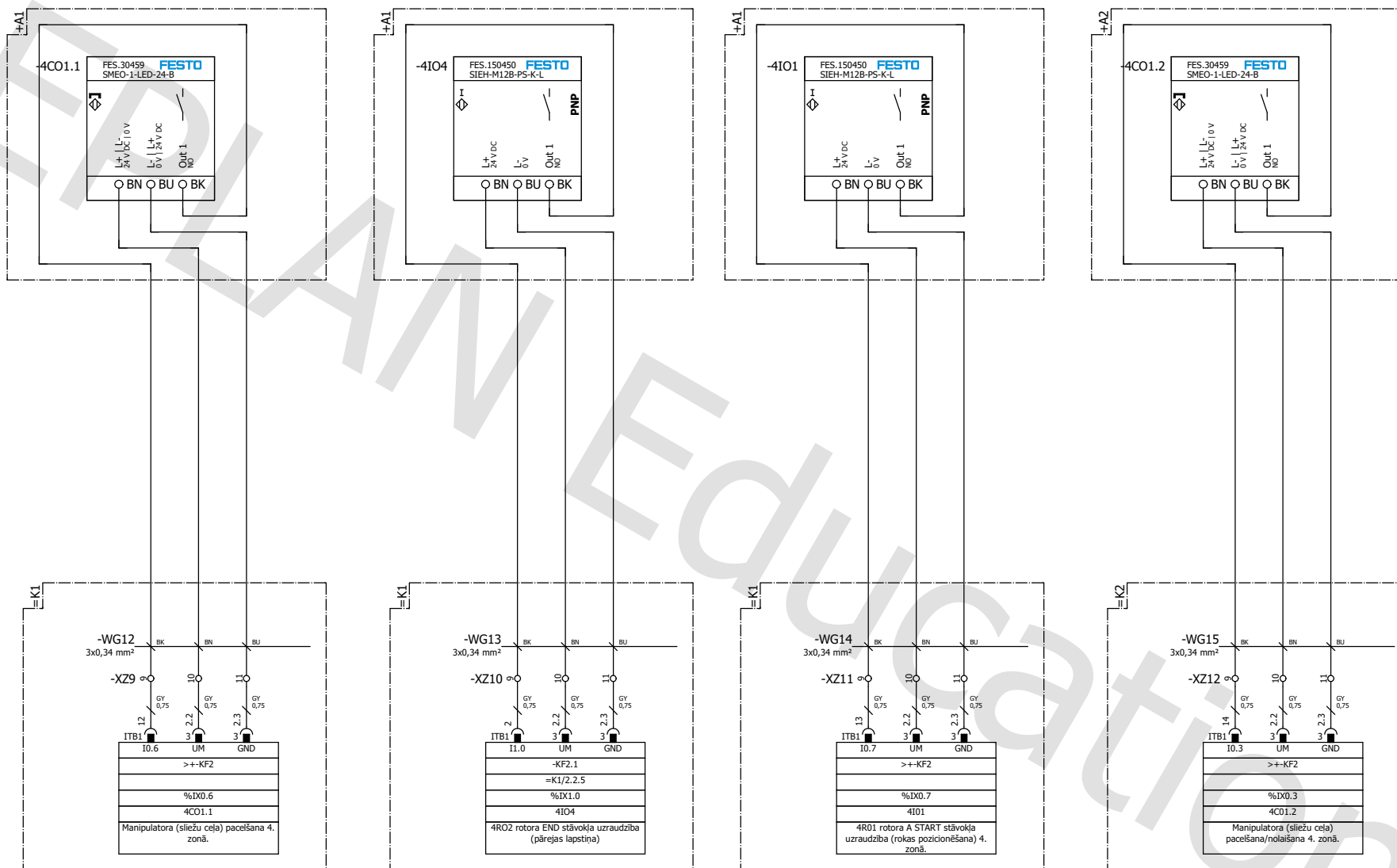
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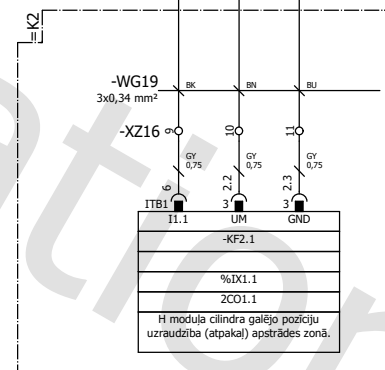
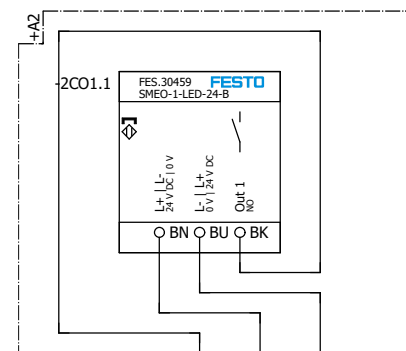
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FESTO station — Loading / Processing / Unloading				Drawing number				Feed workplace: Transport		Electrical engineering schematic	
Creator EPL				Approved by		Date 28.01.2026		Ed. dlno3		Page 20 from 27	





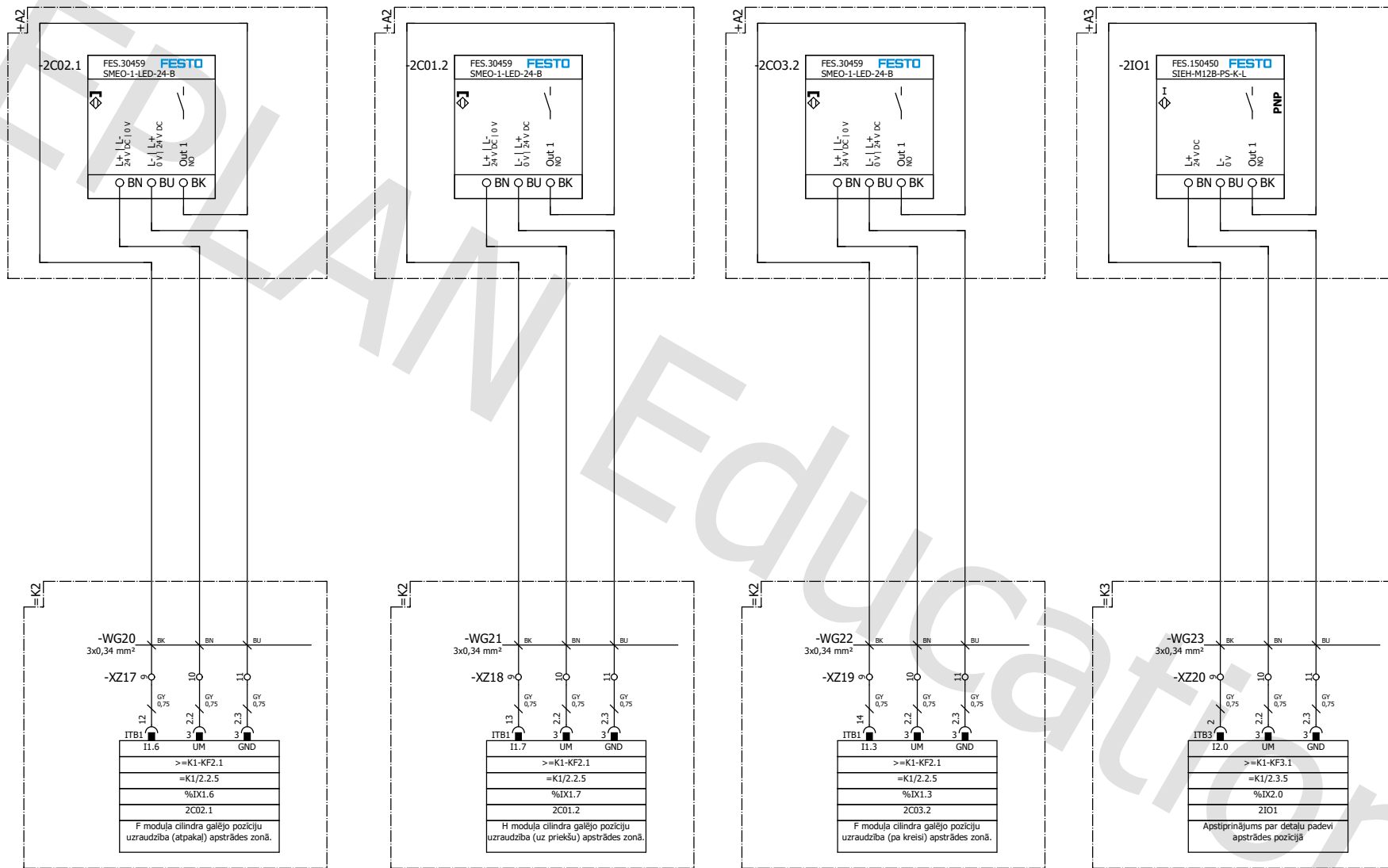
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FESTO station — Loading / Processing / Unloading				Drawing number								Feed workplace: Transport				Electrical engineering schematic			
Creator EPL				Approved by				Date 28.01.2026				Ed. dino3				Page 21 from 27			
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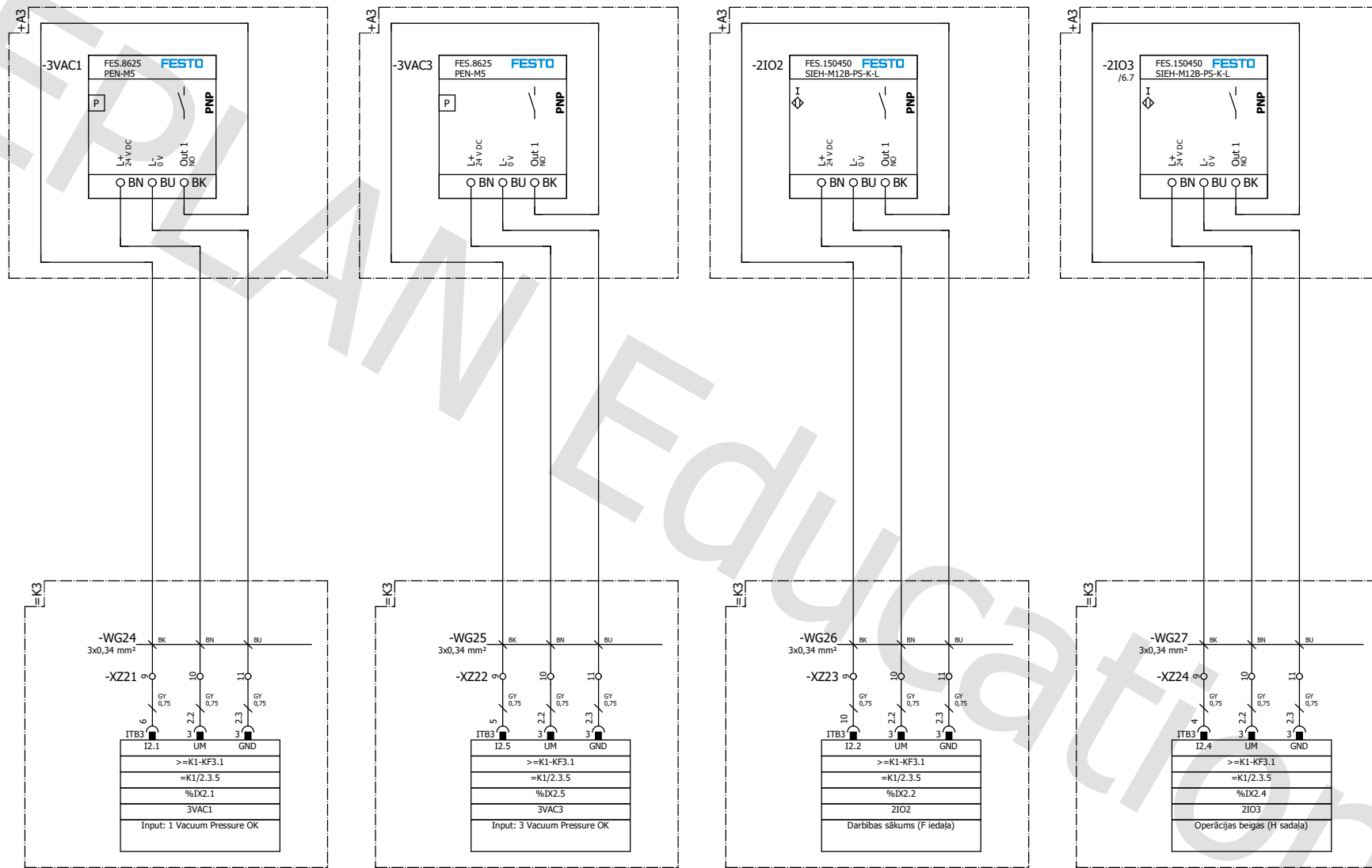


			Project name	Festo Station	Job number		Sensor control				=GL1	&EFS1		
			FESTO station — Loading / Processing / Unloading		RTU-MPS-001						Feed workplace: Transport	Electrical engineering schematic		
					Drawing number						+A2	Page		
Modification	Date	Name	Creator	EPL	Approved by					Date	28.01.2026	Ed.	dino3	Enclosure 2

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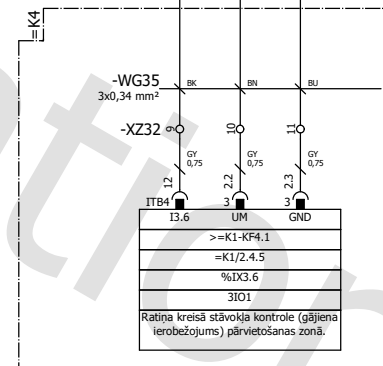
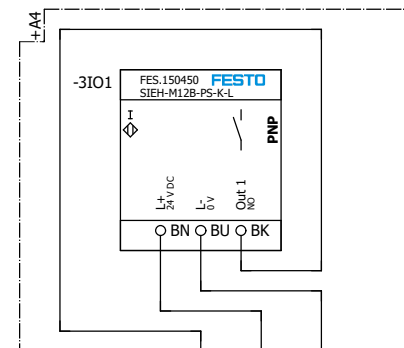
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				FESTO station — Loading / Processing / Unloading	Drawing number	
Modification	Date	Name	Creator	EPL	Approved by	



Sensor control	
Date	28.01.2026
Ed.	dino3

	=GL1	&EFS1
	Feed workplace: Transport	Electrical engineering schematic
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			Project name	Festo Station	Job number		Sensor control			=GL1	&EFS1			
			FESTO station — Loading / Processing / Unloading		RTU-MPS-001					Feed workplace: Transport	Electrical engineering schematic			
					Drawing number						+A2	Page	7	
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